Machine Learning 2 (APM_50441_EP/DSAIB - 2026-2027). PC3

Exercise 1 (Exercise 10.1 from ESL)

AdaBoost is a forward stagewise additive model for the exponential loss: given the current function f_{m-1}, the new classifier $G_m \in \mathcal{G}$ (with values in $\{-1, 1\}$) and its weight β_m minimize $\sum_{i=1}^n e^{-Y_i(f_{m-1}(X_i)+\beta G(X_i))}$, and $f_m = f_{m-1} + \beta_m G_m$. (In the slides, G_m, β_m and err_m are denoted h_t, α_t and ε_t, with weights normalized to sum to 1.) Prove that the optimal step β_m is given by

$$\beta_m = \frac{1}{2} \log \frac{1 - \mathrm{err}_m}{\mathrm{err}_m}$$

where

$$\mathrm{err}_m = \frac{\sum_{i=1}^n w_i^{(m)} I(G_m(X_i) \neq Y_i)}{\sum_{i=1}^n w_i^{(m)}}$$

and $w_i^{(m)} = e^{-Y_i f_{m-1}(X_i)}$ are the weights at step m.

Exercise 2 (Exercise 10.5 from ESL (optional))

For a K-class classification problem with class $V \in \{1, \ldots, K\}$, consider the coding $Y = (Y_1, \ldots, Y_K)^\top$ with

$$Y_k = \begin{cases} 1 & \text{if } V = k \\ -\frac{1}{K-1} & \text{otherwise} \end{cases}$$

Let $f = (f_1, \ldots, f_K)^\top$ with

$$\sum_{k=1}^K f_k = 0$$

and define

$$L(Y, f) = \exp\left(-\frac{1}{K} Y^\top f\right).$$

1. Using Lagrange multipliers, derive the population minimizer f^* of $\mathbb{E}[L(Y, f(X))]$, subject to the zero-sum constraint, and relate these to the class probabilities.
2. Show that a multiclass boosting using this loss function leads to a reweighting algorithm similar to Adaboost, as in Section 10.4.

Exercise 3 (Bagging: averaging correlated predictors)

Let $\hat{f}_1, \ldots, \hat{f}_B$ be predictors (e.g. trees grown on bootstrap samples), evaluated at a fixed point x, identically distributed with variance σ^2 and pairwise correlation $\rho \geq 0$.

1. Prove that the variance of their average $\bar{f} = \frac{1}{B} \sum_{b=1}^B \hat{f}_b$ is $\rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. What happens when $B \to \infty$? How do random forests decrease the variance further?
2. A bootstrap sample draws n observations with replacement among n. Prove that the probability that a given observation is not drawn tends to $1/e \approx 0.368$ when $n \to \infty$. Deduce the principle of the out-of-bag (OOB) estimate.

Exercise 4 (The dual problem of the SVM)

We consider the soft-margin SVM with a feature map ϕ and labels $Y_i \in \{-1, 1\}$:

$$\min_{\beta, \beta_0, \xi} \frac{1}{2}\|\beta\|^2 + C \sum_{i=1}^n \xi_i \quad \text{subject to} \quad Y_i(\beta^\top \phi(X_i) + \beta_0) \geqslant 1 - \xi_i,\ \xi_i \geqslant 0\,.$$

1. Write the Lagrangian with multipliers $\alpha_i \geqslant 0$ (margin constraints) and $\mu_i \geqslant 0$ ($\xi_i \geqslant 0$), and show that minimizing it in (β, β_0, ξ) gives $\beta = \sum_i \alpha_i Y_i \phi(X_i)$, $\sum_i \alpha_i Y_i = 0$ and $\alpha_i = C - \mu_i$.
2. Deduce the dual problem $\max_\alpha \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j Y_i Y_j k(X_i, X_j)$ subject to $0 \leqslant \alpha_i \leqslant C$ and $\sum_i \alpha_i Y_i = 0$, with $k(x, x') = \phi(x)^\top \phi(x')$. Why does it only involve the kernel?
3. Using the complementary slackness conditions $\alpha_i \left(Y_i f(X_i) - 1 + \xi_i\right) = 0$ and $\mu_i \xi_i = 0$, show that $\alpha_i = 0$ for the observations strictly outside the margin ($Y_i f(X_i) > 1$), $\alpha_i = C$ for those strictly inside ($Y_i f(X_i) < 1$), and $0 \leqslant \alpha_i \leqslant C$ on the margin. Which observations determine the classifier?

Exercise 5 (Random Fourier features)

Let $\gamma > 0$, $W \sim \mathcal{N}(0, 2\gamma I_d)$ and $b \sim \mathcal{U}[0, 2\pi]$ independent, and $z(x) = \sqrt{2}\cos(W^\top x + b)$ for $x \in \mathbb{R}^d$.

1. Prove that $\mathbb{E}[z(x)z(x')] = \mathbb{E}\big[\cos(W^\top(x - x'))\big]$. (Use $2\cos a \cos b = \cos(a-b) + \cos(a+b)$.)
2. Using the characteristic function of a Gaussian vector, $\mathbb{E}\left[e^{iW^\top u}\right] = e^{-\gamma\|u\|^2}$, deduce that $\mathbb{E}[z(x)z(x')] = e^{-\gamma\|x-x'\|^2}$, the RBF kernel. How can a kernel method be approximated by a linear one, and what is the gain?

Exercise 6 (XGBoost: second-order boosting)

At each step, XGBoost adds a tree h with T leaves of values $w_1, \dots, w_T$ to the current score F, minimizing a second-order approximation of the loss plus a penalty:

$$\sum_{i=1}^{n} \left(g_i h(x_i) + \frac{1}{2} H_i h(x_i)^2 \right) + \gamma T + \frac{\lambda}{2} \sum_{j=1}^{T} w_j^2 \,,$$

where g_i and H_i are the first and second derivatives of the loss $\ell(y_i, \cdot)$ at $F(x_i)$.

1. For a fixed partition into leaves $I_1, \dots, I_T$, with $G_j = \sum_{i \in I_j} g_i$ and $H_j = \sum_{i \in I_j} H_i$, show that the optimal leaf values are $w_j^\star = -G_j/(H_j + \lambda)$ and that the optimal objective is $-\frac{1}{2}\sum_j \frac{G_j^2}{H_j + \lambda} + \gamma T$.
2. Deduce the gain of splitting a leaf into a left and a right part, and interpret γ and λ. For the logistic loss $\ell(y, F) = \log(1 + e^{-yF})$ with $y \in \{-1, 1\}$, compute g_i and H_i.

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